

Apostol, Chapter 1 - Exercise 20

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20. Let $d = (826, 1890)$. Use the Euclidean algorithm to compute d , then express d as a linear combination of 826 and 1890.

$$1890 = 826 \times 2 + 238$$

$$826 = 238 \times 3 + 112$$

$$238 = 112 \times 2 + 14$$

$$112 = 14 \times 8 + 0,$$

then $d = 14$. Furthermore, using the equations above we conclude that,

$$14 = 7 \times 1890 - 16 \times 826.$$